Wireless Networking

Tasks

1.List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

ANS:

Easy Explanation

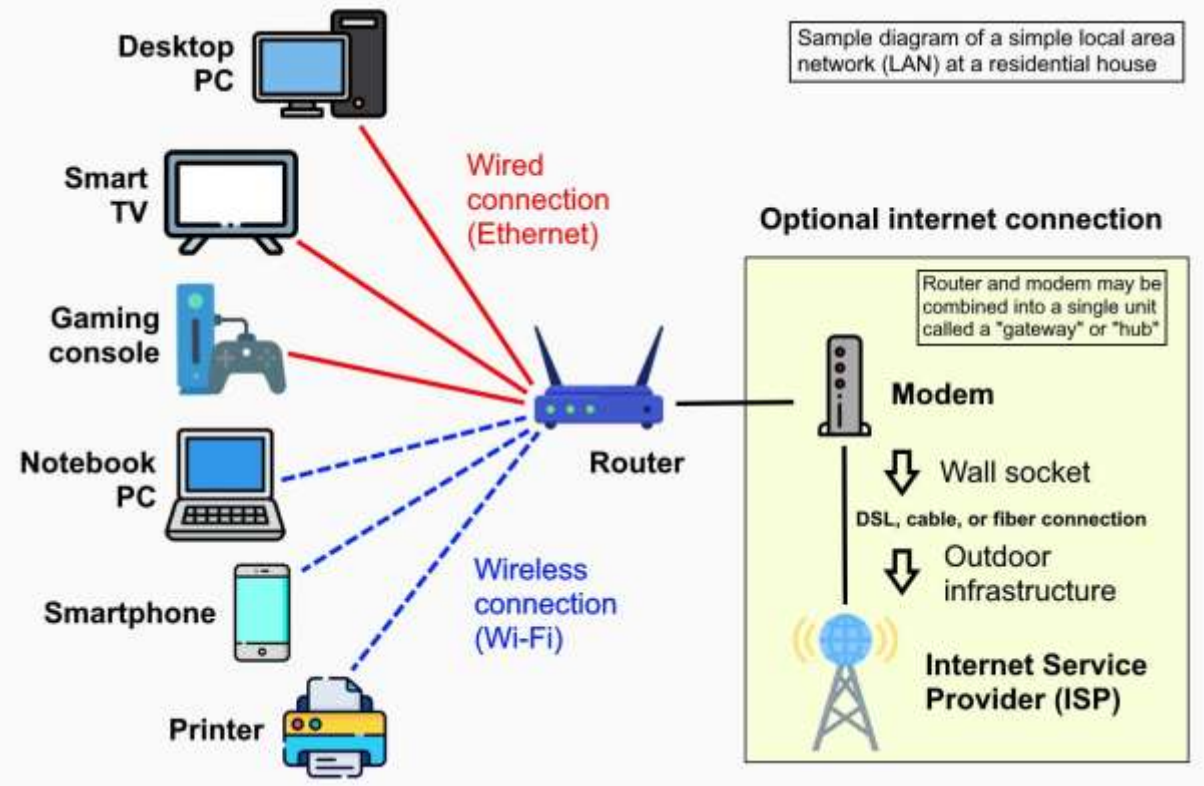
1. 802.11n: Good for normal internet use.
2. 802.11ac: Faster for video streaming and gaming.
3. 802.11ax: Faster and better when many devices connect together.

Real-Life Use

- Wi-Fi 4: Basic home internet.
- Wi-Fi 5: Watching HD videos.
- Wi-Fi 6: Online gaming and connecting many devices.

2.Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

ANS:



1. Smartphone connects to the Wi-Fi router wirelessly.
2. Router connects to the modem using a cable.
3. Modem connects to the internet.
4. Smartphone can now browse websites, watch videos and use apps.

Network Parts

Connection	Type
Smartphone to Router	Wireless (Wi-Fi)
Router to Modem	Wired (Ethernet cable)
Modem to Internet	Wired connection

Real-Life Use

At home, we use Wi-Fi to connect our mobile phones, laptops and smart TVs to the internet.

3.Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

ANS:

Step 1: Open Wi-Fi Settings

1. Open your laptop.
2. Click the Wi-Fi icon.
3. Check the available Wi-Fi networks.
4. Note their names and signal strength.

Step 2: Wi-Fi Network Details

Sample observation table (example only):

No.	Wi-Fi Name (SSID)	Signal Strength	Security
1	JioFiber_123	Strong	WPA2
2	Airtel_WiFi	Medium	WPA3
3	Home_Network	Weak	WPA2

Step 3: Which Wi-Fi Would I Choose?

I would choose Airtel_WiFi because:

1. It has WPA3 security.
2. It provides a good signal.
3. It helps protect my internet connection.

Real-Life Use

We check Wi-Fi names and signal strength before connecting our phones and laptops to the internet.

4.Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

ANS:

Meaning:

SSID hiding means hiding the Wi-Fi network name so it does not appear in the list of available Wi-Fi networks.

Advantage:

- Other people cannot easily see your Wi-Fi name.

Disadvantage:

- Users must enter the Wi-Fi name manually to connect.

Real-Life Example

At home, we can hide our Wi-Fi name so it does not appear in the Wi-Fi list on other phones.